RESEARCH ANALYSIS FIELD GUIDE

Statistical tests + Machine Learning + Deep Learning | Research scenarios and method selection

1 | Start with the research objective

Objective	Typical question	Suitable methods
Describe	What are the demographics and score distributions?	Frequencies, percentages, mean/median, SD, plots
Compare	Do two groups or multiple conditions differ?	t-test, ANOVA, paired tests, rank-based alternatives
Associate	Are variables related?	Pearson/Spearman correlation, chi-square
Explain / estimate	What is the adjusted effect of X on Y?	Linear/logistic regression, multilevel models, SEM
Predict	Can we predict an unseen patient or student outcome?	Regularized regression, random forest, gradient boosting, neural nets
Discover	Are there subgroups or latent dimensions?	PCA, factor analysis, k-means, hierarchical clustering

2 | Statistical test selector for real studies

Research scenario	Recommended analysis	Check / report
Two independent groups: intervention vs control, numeric score	Welch independent t-test; Mann-Whitney for rank/distribution questions	Independence; distribution/outliers; effect size and CI
Before/after on same participants	Paired t-test; Wilcoxon signed-rank when appropriate	Paired differences; CI; within-person design
Three+ independent groups	Welch ANOVA; one-way ANOVA; Kruskal-Wallis	Variance assumptions; post-hoc correction; effect size
Three+ repeated measurements	Repeated-measures ANOVA or mixed-effects model; Friedman alternative	Within-person dependence; missing data; time effects
Two categorical variables (e.g., gender x adoption)	Chi-square independence; Fisher exact for sparse tables	Expected cell counts; odds/risk measures when applicable
Two continuous variables	Pearson correlation; Spearman for monotonic/rank association	Scatterplot; outliers; correlation CI

Research scenario	Recommended analysis	Check / report
Continuous outcome with multiple predictors	Multiple linear regression; mixed-effects for clustered observations	Linearity, residuals, collinearity; coefficients and CI
Binary outcome (pass/fail, disease/no disease)	Logistic regression	Calibration; odds ratios and CI; class imbalance
Time to event (e.g., relapse)	Kaplan-Meier, log-rank, Cox regression	Censoring; proportional hazards for Cox
Questionnaire construct validity	EFA / CFA; reliability omega or alpha	Dimensionality, item quality, model fit, sample adequacy

3 | ML and DL options by data modality

Data and scenario	Practical baseline / ML	DL option (when justified)	Evaluation
Survey / tabular: predict student dropout	Logistic regression, random forest, XGBoost	Tabular MLP / transformer with adequate data	Stratified CV; ROC-AUC, PR-AUC, calibration
EEG / ECG: classify signal patterns	Engineered features + SVM / boosted trees	1D CNN, temporal CNN, transformer	Subject-level split; sensitivity, specificity
Images: detect lesions or segment objects	Radiomics + SVM; classical image processing	CNN, U-Net, vision transformer	Patient-level split; Dice/IoU for masks
Video: recognize continuous sign language	Pose keypoints + HMM / sequence classifier	Spatiotemporal CNN, transformer, CTC-based CSLR	Signer-independent split; word error rate
Text: classify feedback / sentiment	TF-IDF + logistic regression / SVM	Fine-tuned language model	Macro-F1; error analysis
Spatial-temporal: forecast regional patterns	Lag features + boosting / statistical time series	Temporal GNN, ConvLSTM, transformer	Forward-time / geographic holdout

4 | Decision and quality checklist

- Define outcome, unit of analysis, estimand and study design before choosing a method.
- Choose tests by outcome scale, independence/paired structure, groups, assumptions and sample adequacy.
- Use effect sizes, uncertainty intervals and corrected multiplicity where relevant; do not rely only on p-values.
- For ML/DL, establish a simple baseline first; separate train/validation/test and avoid leakage.
- Split by participant, site or time when deployment requires generalization to new participants, sites or periods.
- For imbalanced outcomes, inspect precision-recall, sensitivity, specificity and calibration, not accuracy alone.
- Prediction does not establish causation; observational regression also needs a credible identification strategy for causal claims.

• Report missing-data handling, preprocessing, hyperparameters, uncertainty, fairness and external validation.

5 | References and reading

OpenIntro Statistics (2019): https://www.biostat.jhsph.edu/~iruczins/teaching/books/2019.openintro.statistics.pdf
Naked Statistics: https://github.com/Abcsunny95/Data-Science-Books/blob/master/Naked%20Statistics.pdf
Statistics for Dummies: https://www.mymathscloud.com/api/download/modules/AP-Statistics/Textbooks/Statistics%20for%20Dummies.pdf?id=25302635
The Cartoon Guide to Statistics: https://cdn.schools.in/DIG_LIB/E-Book/The%20cartoon%20guide%20to%20statistics%20(2).pdf

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